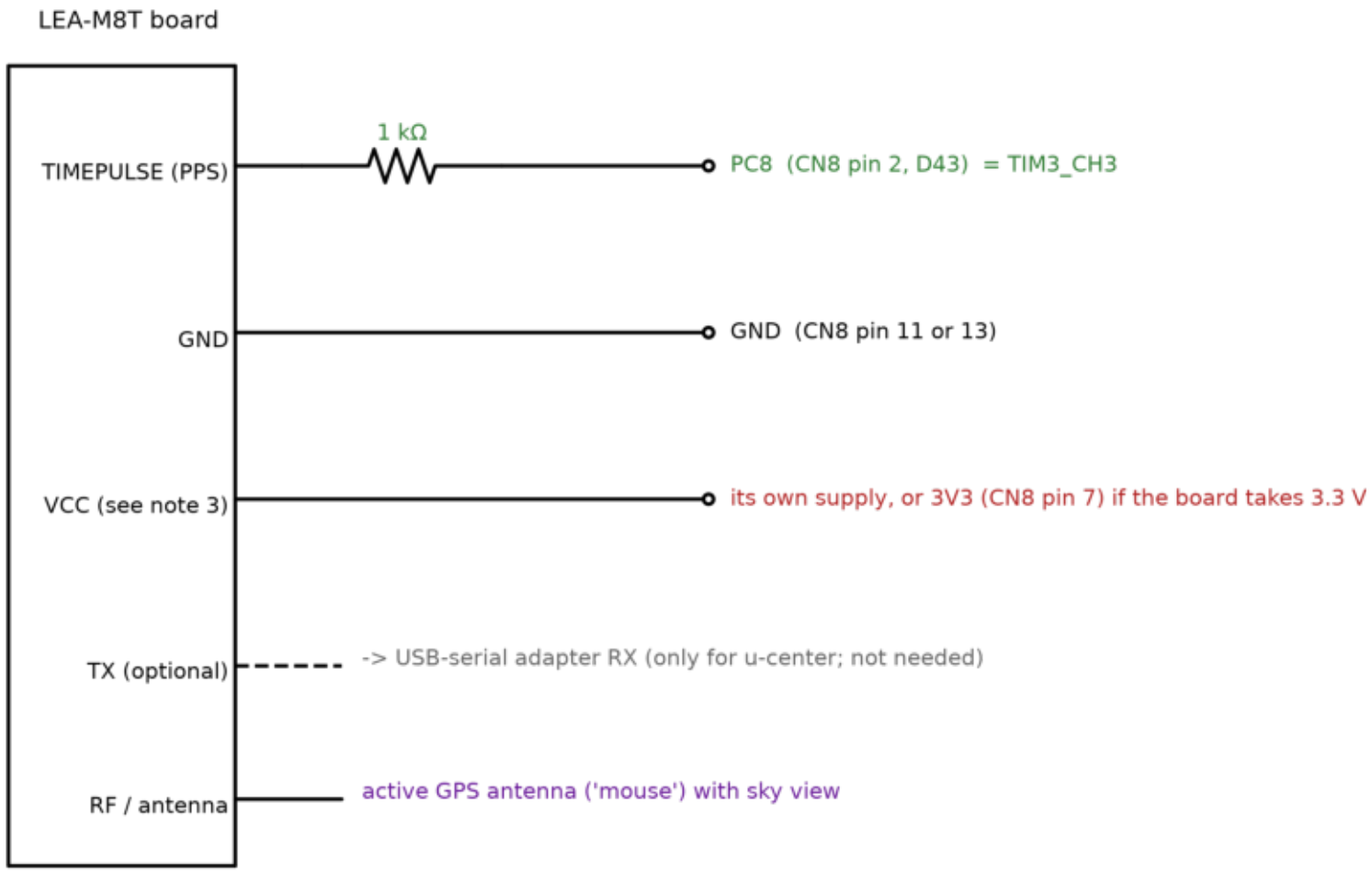
GPS PPS for the one-off calibration: LEA-M8T -> NUCLEO-H723ZG (3 wires + one 1 kΩ)

Firmware /data/claude/nucleo-h723/gps/rx_build_ocxo_gps.bin: the pulse is captured on PC8 = TIM3_CH3, the counter that timestamps the OCXO and the ADC samples (5



1. Power order as with the OCXO: NUCLEO on USB first, then the GPS board; off in the reverse order. Common GND is required.
2. TIMEPULSE -> 1 kΩ -> PC8. The 1 kΩ protects the pin whatever the board's output level is (3.3 V module pin, or 5 V after a level shifter).
   PC8 is on connector CN8, pin 2 (marked D43 / PC8) - please check the marking on your board before connecting. The firmware sets a pull-down there.
3. Supply: LEA-M8T modules take 2.7-3.6 V; many breakout boards have their own regulator and take 5 V - use what the board says.
   From the NUCLEO: 3V3 = CN8 pin 7 (only if the board is a 3.3 V board), 5V = CN8 pin 9. The GPS draws ~30-70 mA plus the active antenna.
4. No configuration needed: the default TIMEPULSE is 1 pulse per second (rising edge = the second). GPS time and UTC differ by exactly 18 s,
   so the fraction of the second is the same. Antenna with a clear view of the sky; wait for a fix (the board's LED blinks with the PPS).
5. Then the new firmware is flashed; from the first pulse the log shows one 'GPS PPS' line per second (OCXO vs GPS, e-CzasPL clock and DCF77 second
   vs the GPS second). Leave it connected ~24 h (day and night); tools/gps_calib.py turns the lines into the calibration.